Chapter 10, Problem 5RQ

Problem

The current *i* in the circuit of Fig. 3.48 is:

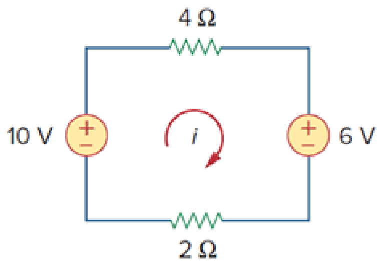
(a) −2.667 A

(b) −0.667 A

(c) 0.667 A

(d) 2.667 A

Figure 3.48



Step-by-step solution

Step 1 of 2

If a circuit contains sources of different frequencies, then we have to find the current i_x due to the source $\sin(2t)$ individually and the current i_x due to the source $\sin(10t)$ individually then add both the responses.

Step 2 of 2

So, when the sources are different frequencies then we have to use the superposition theorem.

Therefore, the correct option id $\boxed{(b)}$.